Prompting Strategies Do Not Transfer: Model-Dependent Technique Effects for Multimodal Forensic Video Analysis

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Abstract

Prompting strategies are typically evaluated on a single model and reported as properties of the strategy. We test that generalisation for forensic video analysis, evaluating eight strategies (zero-shot, sequential, chain-of-thought, least-to-most, ReAct, meta-prompting, self-consistency, and iterative refinement) on UCF-Crime across three frontier multimodal models, with a uniform three-judge panel labelling 19,361 model–technique–video cells.

No strategy was optimal across models. Zero-shot ranked first for GPT-5.5 (0.596) and seventh of eight for Claude Opus 4.7 (0.498); least-to-most ranked first for Claude (0.616) and worst for GPT-5.5 (0.096). Variation across strategies within a model (17–50 accuracy points) far exceeded variation across models (4.4 points under a matched output schema).

Decomposing accuracy into commitment rate, meaning how often a model asserts a crime occurred, and conditional accuracy is informative: commitment rate tracked accuracy at $r = 0.96$ and 0.81, while conditional accuracy stayed within 0.660–0.694 across GPT-5.5's five mainstream strategies and was orthogonal to accuracy under Claude. These patterns are consistent with prompting acting primarily on how readily a model asserts a crime, which would explain why an identical scaffold helps one model and harms another.

Two deployment findings follow. Stated confidence was anti-correlated with correctness in two of three models. And in justification text, markers of social status were associated with elevated error (RR 1.21, 1.39) while markers of concealment were associated with reduced error (RR 0.59, 0.76). No confidence signal exposes the asymmetry.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; **Video analysis**; • **General and reference** → *Evaluation*; • **Social and professional topics** → *Computing / technology policy*.

Keywords

multimodal large language models, prompting strategies, forensic video analysis, evaluation, calibration, abstention, UCF-Crime

1 Introduction

Multimodal large language models (MLLMs) are increasingly proposed as triage aids for forensic video analysis, where expanding surveillance capture has outpaced analytic capacity [1, 2]. Unlike detection-oriented vision models, MLLMs produce structured narrative reasoning that resembles investigative practice, which makes the choice of *prompting strategy* a first-order design decision rather than an implementation detail [3, 4].

A substantial literature reports comparative evaluations of such strategies. Chain-of-thought [3], least-to-most [5], ReAct [4], self-consistency [6], and meta-prompting [7] are each introduced with evidence of improvement over direct prompting, and downstream studies rank them against one another. Almost universally, these comparisons are conducted on a single model, and their conclusions are stated as properties of the *strategy*: chain-of-thought elicits reasoning; self-consistency improves it; ReAct grounds it. Practitioners reasonably read such claims as portable guidance.

This paper asks whether that portability holds. If a strategy's advantage is a property of the strategy, it should reproduce across independently developed model lineages of comparable capability. If instead the advantage is a property of the *strategy–model pair*, then single-model comparisons license no recommendation at all, and a body of published guidance is being applied outside its evidential scope.

The question is not merely methodological in a forensic setting. A triage system that declines to assert a crime produces missed incidents; one that asserts too readily produces false accusations against people who appear in surveillance footage. These are different harms falling on different parties, and we show that prompting structure, a choice usually made on grounds of convenience or convention, moves a system between them.

We evaluate eight prompting strategies on UCF-Crime [8] across three frontier MLLMs from independent developers, scoring 19,361 model–technique–video cells. We ask three research questions.

- **RQ1** Do prompting-strategy rankings transfer across model families, or is strategy effectiveness model-dependent?
- **RQ2** If effects are model-dependent, what does a prompting scaffold actually change: perceptual discrimination, or the threshold at which a model commits to an assertion?
- **RQ3** What follows for forensic deployment, given that practitioners must choose a strategy, trust a confidence signal, and accept a residual error profile?

We test this on one task where the cost of a misleading recommendation is high; Section 6 discusses what the result does and does not license beyond it.

Our contributions are: (i) a cross-family test of whether prompting-strategy rankings transfer, covering eight strategies, three independently developed models and the full UCF-Crime test partition; (ii) a decomposition of accuracy into a commitment component and a discrimination component, which separates what a scaffold changes about a model's willingness to answer from what it changes about the model's ability to answer correctly; (iii) an assessment of whether the confidence these systems report, and the agreement across their sampled generations, can be used as trust signals;

(iv) an analysis of which crime categories resist every strategy, distinguishing limits of the reasoning procedure from limits of the input pipeline; and (v) an audit of the descriptors these systems cite as supporting evidence, asking which classes of descriptor are associated with correct and with incorrect classification.

We evaluate these strategies as candidate methods for forensic workflows. Evidentiary validation (chain of custody, authentication, admissibility) is outside the present scope and is discussed in Section 6.

2 Related Work

Prompting strategies. Chain-of-thought prompting elicits intermediate reasoning steps and improves performance on multi-step tasks [3]; zero-shot variants achieve similar effects without exemplars [9]. Least-to-most decomposes a problem into progressively harder sub-problems [5]. Self-consistency samples multiple reasoning paths and aggregates by majority [6]. ReAct interleaves reasoning with actions and observations [4]. Meta-prompting has the model generate its own task prompt [7]. Each is introduced with single-model or single-family evidence, and subsequent comparative studies inherit that design. Our contribution is orthogonal to any individual strategy: we test whether the comparative conclusions themselves generalise.

MLLMs for video and forensic analysis. Contemporary MLLMs process visual input while producing analytic narrative [10–12]. Applications to media forensics have examined deepfake detection [2] and image enhancement [1], while forgery analysis has been addressed with specialised architectures [13]. Our prior work comparing prompting strategies for forensic video analysis scored outputs against a ten-criterion forensic rubric and reported ReAct as the strongest strategy for all three models tested [14]. That study measured narrative quality on 22 clips; the present work measures classification correctness on 807, and finds a different ordering, a divergence we return to in Section 5.

Anomaly detection benchmarks. UCF-Crime [8] is the standard real-world surveillance anomaly benchmark, providing untrimmed videos across thirteen anomaly categories. The UCA extension supplies timestamped sentence-level event descriptions [15]. Most work treats the task as weakly supervised detection; we treat it as closed-set classification from sampled frames, which is the operating mode of an MLLM triage aid.

Calibration and abstention. Verbalised confidence from language models is known to be poorly calibrated [16, 17], and selective prediction frameworks trade coverage against risk [18]. Abstention has been studied primarily as a safety behaviour; we treat it as the dependent variable that prompting strategies actually manipulate, which to our knowledge has not been reported for multimodal video analysis.

Algorithmic decision-making in policing contexts. Automated risk and surveillance tools have documented disparate effects on overpoliced communities [19, 20]. Our concern is adjacent but distinct: we show that a purely technical parameter, prompt structure, determines the operating point between missed detections and false accusations, and that this choice is currently made implicitly.

3 Method

3.1 Dataset

We use the UCF-Crime test partition [8], 812 videos spanning eleven anomaly categories: RoadAccidents (150), Robbery (150), Burglary (100), Stealing (100), Abuse (50), Explosion (50), Fighting (50), Shooting (50), Shoplifting (50), Vandalism (50), and Assault (12). Frames were sampled uniformly per video (median 91 frames) and submitted in batches. Reference descriptions for judge-based scoring come from the UCA sentence-level annotations [15] (median 6 sentences per video).

The evaluated label set contains no *Normal* videos, so a response asserting a normal scene is always incorrect. We therefore report classification among anomaly categories, not anomaly detection; consequences are discussed in Section 6.

Two sample sizes appear throughout. Direct parsing (Section 3.4) uses the full 812-video partition. The judge panel (Section 3.5) covers the 807 of those videos for which a UCA reference and a complete pair of judge labels were both available, giving $807 \times 8 \times 3 = 19{,}368$ possible cells, of which 19,361 were scored. Individual cells fall below these totals where a run was truncated or a response could not be parsed; per-cell n is reported in every table.

3.2 Models

Three frontier MLLMs from independent developers: GPT-5.5, Claude Opus 4.7, and Gemini 3.1 Pro.[1] We use *model family* to mean an independently developed model lineage, as distinct from versions within a lineage.

3.3 Prompting strategies

Eight strategies were implemented: **zero-shot** (direct instruction); **sequential** (staged progressive analysis); **chain-of-thought** (explicit reasoning steps); **least-to-most** (graded sub-problem decomposition); **ReAct** (thought–action–observation cycles); **meta-prompting** (model-generated task prompt); **self-consistency** (n=5 samples, majority aggregation); and **iterative refinement** (repeated revision under a similarity-based stopping rule). No few-shot exemplars were used in any condition.

3.4 Outcome measures

For GPT-5.5 and Claude Opus 4.7 the harness elicited a structured verdict (classification, confidence, severity, supporting evidence), which we parse directly. We report:

- **Accuracy** and **macro-F1** against the ground-truth category.
- **Commitment rate** CR: the proportion of videos for which the model asserts some crime category, rather than reporting a normal scene or abstaining.
- **Conditional accuracy** CA: accuracy restricted to committed responses. Since every *Normal* response is incorrect on this partition, the decomposition

$$\text{Acc} = \underbrace{\text{CR}}_{\text{threshold}} \times \underbrace{\text{CA}}_{\text{discrimination}} \tag{1}$$

[1] Exact API model strings, temperatures, batch sizes, and token limits are available from the authors on request. UCF-Crime videos are available from the original dataset release [8] under its own terms; we redistribute no videos or frames.

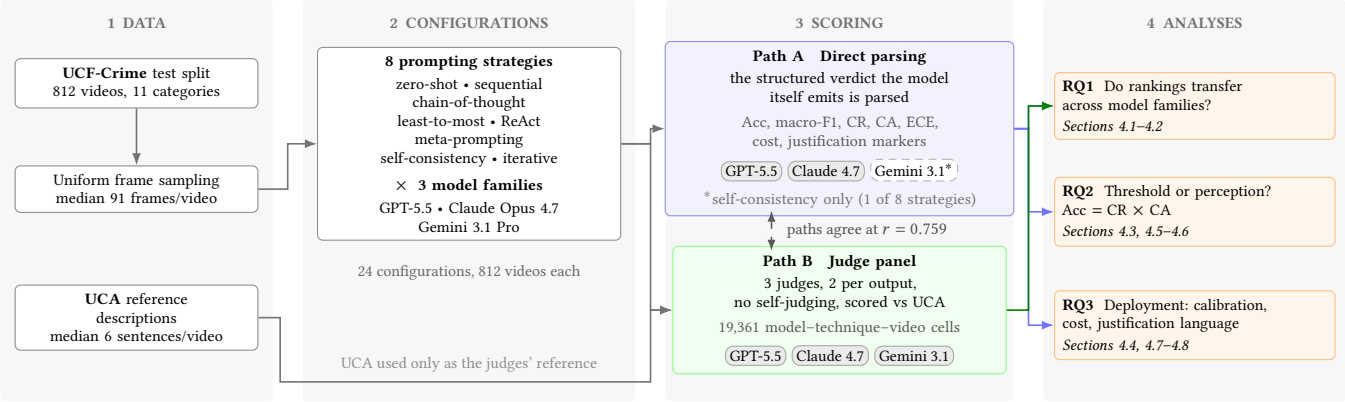


Figure 1: Experimental design. Every strategy is applied to the same 812 videos under all three models. Path A parses the structured verdict the model emits and supports the full metric set, including the exact decomposition Acc = CR × CA; it covers GPT-5.5 and Claude, because the Gemini harness elicited prose rather than a labelled verdict for seven of the eight strategies. Path B applies a single judge rubric to all three models at coarser resolution and so carries the three-model comparison. The two paths agree at $r = 0.759$ over the fourteen cells where both are available, so Path A is treated as primary and Path B as corroboration.

holds exactly, separating technique effects on a model's willingness to assert a crime from its ability to name the right one.

- **Calibration**: expected calibration error (10 bins), mean confidence minus accuracy, and the point-biserial correlation between stated confidence and correctness.
- **Justification markers**: presence, in the model's stated supporting evidence, of (a) *status* descriptors (homelessness, vagrancy, panhandling, dishevelment, intoxication, loitering); (b) *apparel* descriptors (hooded or masked figures, obscured faces); and (c) explicit demographic descriptors. Markers are detected by person-anchored regular expressions; we report the relative risk RR = $P(\text{error} \mid \text{marker})/P(\text{error} \mid \text{no marker})$ with χ^2 tests and 95% intervals.
- **Cost**: mean generated tokens per video, and tokens per correct classification.

3.5 Uniform panel labelling

The Gemini harness elicited prose rather than a structured verdict for seven of the eight strategies; only its self-consistency run used the shared schema, which is why Gemini appears in the calibration comparison of Section 4.4 but not in Tables 1 and 2. To obtain a common outcome measure across all three models we use a three-judge panel (GPT, Claude, Gemini as judges), each output scored by two judges under an identical rubric against the UCA reference. Self-judging was excluded by design. A cell counts as correct when the panel records neither a missed crime nor a misclassification.

Excluding self-judging has a consequence for interpretation: each subject model is assessed by a *different* pair of judges, and the judges differ in severity. Panel scores are therefore not directly comparable *between* models, and we do not use them to rank the models against one another. We use them for the comparison they do support, namely the ordering of strategies *within* each model, where the judge pair is held constant.

Table 1: GPT-5.5. CR = commitment rate; CA = conditional accuracy.

Strategy	Acc	Macro-F1	CR	CA	Normal	*n*
Zero-shot	**0.596**	0.565	0.903	0.660	0.097	812
Sequential	0.482	0.495	0.724	0.665	0.276	812
Self-consistency	0.463	0.499	0.667	0.694	0.333	812
Meta-prompting	0.443	0.475	0.656	0.675	0.344	806
ReAct	0.420	0.441	0.633	0.663	0.367	812
Iterative	0.358	0.369	0.690	0.520	0.306	742
Least-to-most	0.096	0.123	0.271	0.355	0.727	812
Chain-of-thought†	0.000	0.000	0.000	n/a	0.022	811

Panel-derived accuracy agrees with direct label comparison at $r = 0.759$ over the fourteen GPT/Claude cells where both are available (mean absolute difference 0.078). We therefore treat direct comparison as primary and panel labelling as corroborating evidence, not as an interchangeable measurement.

3.6 Statistical analysis

Paired model comparisons use McNemar's exact test; three-way comparison uses Cochran's Q. Rank agreement across strategies uses Spearman's ρ. Given only eight strategies, rank correlations have low power; we therefore report specific rank inversions as the substantive evidence and treat ρ as descriptive.

4 Results

4.1 No strategy is optimal across models (RQ1)

Figure 1 summarises the design. Tables 1 and 2 report performance under direct label comparison.

† Pipeline defect: the classification stage did not receive the preceding analysis stage's output, producing abstention in 99.9% of cases. Excluded from all subsequent analysis. Chain-of-thought under

Table 2: Claude Opus 4.7. Coverage varies where API rate limiting truncated a run.

Strategy	Acc	Macro-F1	CR	CA	Normal	*n*
Least-to-most‡	**0.616**	0.549	0.922	0.668	0.078	625
Sequential	0.599	0.536	0.953	0.628	0.047	793
Chain-of-thought	0.527	0.506	0.851	0.619	0.149	812
ReAct	0.523	0.505	0.868	0.603	0.132	812
Meta-prompting	0.514	0.490	0.851	0.603	0.149	812
Self-consistency	0.498	0.455	0.760	0.655	0.240	717
Zero-shot	0.498	0.459	0.910	0.547	0.090	812
Iterative	0.444	0.508	0.634	0.701	0.366	432

Table 3: Panel-derived accuracy, 807 videos, three models.

Strategy	Claude	GPT-5.5	Gemini
Sequential	**0.486**	0.481	**0.482**
Chain-of-thought	0.462	n/a†	0.472
Zero-shot	0.458	**0.611**	0.411
Meta-prompting	0.440	0.390	0.223
ReAct	0.410	0.436	0.332
Least-to-most	0.371	0.130	0.377
Self-consistency	0.347	0.410	0.426
Iterative	0.252	0.405	0.001

Claude (0.527) confirms this is a harness failure, not a property of the strategy.

‡ Reduced coverage from API rate limiting, missing-not-at-random (all *Stealing*, all *Vandalism*, 36 *Shoplifting*). On the 520-video subset where all strategies completed, least-to-most attains 0.612 against zero-shot's 0.529, preserving the ordering.

The orderings are close to opposed. Zero-shot is best for GPT-5.5 and seventh of eight for Claude. Least-to-most is best for Claude and worst for GPT-5.5, a 0.520 accuracy gap on identical prompts and videos. Spearman rank correlation between the two models is $\rho = -0.25$ ($p = 0.59$, $n = 7$): negative in direction but far from significant at this number of strategies. We therefore rest the claim on the specific inversions, which are large and unambiguous, rather than on the coefficient.

Within-model spread is 0.500 for GPT-5.5 and 0.172 for Claude. Under self-consistency, the one strategy for which all three models used an identical output schema (Section 4.4), the between-model spread is 0.044. These two quantities are not measured on comparable footings: the within-model spread is taken over seven or eight strategies, whereas the between-model spread rests on a single matched strategy and so gives no estimate of how much models differ under other strategies. The comparison indicates that strategy choice can matter at least as much as model choice in this setting; it does not establish that it generally dominates, and a properly matched three-model design across all eight strategies would be needed to settle that.

4.2 Three-model confirmation under uniform labelling

Table 3 reports panel-derived accuracy for all three models.

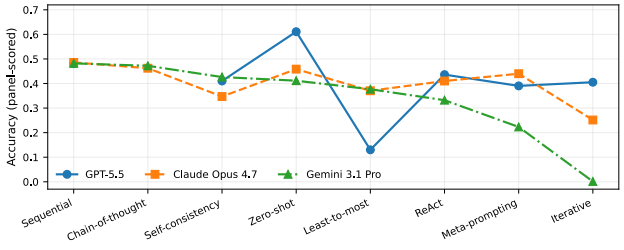


Figure 2: Technique rankings under panel scoring, which applies an identical rubric to all three models. Lines cross repeatedly: sequential prompting leads for Claude and Gemini, zero-shot for GPT-5.5, and self-consistency moves from third for Gemini to seventh for Claude. No strategy is best everywhere.

Figure 2 plots these rankings. Read as within-model orderings, again no strategy leads for all three models: zero-shot for GPT-5.5, sequential for Claude and Gemini. Rank movements are large: iterative refinement ranks fifth for GPT-5.5 and last for both other models; self-consistency ranks third for Gemini and seventh for Claude. Agreement with direct comparison is close for GPT-5.5 (0.611 against 0.596 for zero-shot; 0.481 against 0.482 for sequential) and systematically lower for Claude, whose outputs were assessed by the most severe judge in the panel. This is the expected consequence of the unbalanced judge assignment described in Section 3.5, and it is why we treat Table 3 as evidence that no strategy is universally best rather than as a set of comparable performance estimates across models.

4.3 Scaffolds move the commitment threshold (RQ2)

Applying the decomposition of Equation (1), commitment rate tracks technique-level accuracy closely in both models: $r(\text{Acc}, \text{CR}) = 0.964$ for GPT-5.5 and 0.806 for Claude. Conditional accuracy behaves differently between them. Under Claude it is effectively orthogonal to accuracy ($r = -0.019$): the eight strategies span 0.172 in accuracy while conditional accuracy varies non-monotonically over 0.547–0.701. Under GPT-5.5, conditional accuracy lies within 0.660–0.694 for the five mainstream strategies (zero-shot, sequential, self-consistency, meta-prompting, and ReAct) while accuracy over the same five spans 0.420–0.596. The two remaining strategies break the pattern: least-to-most and iterative refinement have lower conditional accuracy as well as lower commitment (0.355 and 0.520 against a 0.660–0.694 baseline), so across all seven strategies conditional accuracy also correlates with accuracy for GPT-5.5 ($r = 0.907$).

These observations are *consistent with* prompting acting primarily on how readily a model asserts a crime, with a minority of scaffolds additionally disturbing the analysis that precedes the assertion. We stop short of a causal claim. The decomposition is an accounting identity, not an experiment: it partitions accuracy into two observable quantities but does not establish that a scaffold operates on one of them. Nor should conditional accuracy be read as a direct measure of perception; it is accuracy among committed

Table 4: Calibration under self-consistency, the harness-matched strategy. Each row is computed over that model's parsed responses, so accuracies match Tables 1–2 ($n = 812$ for GPT-5.5, 717 for Claude); Gemini is scored on the 717 videos common to all three models.

Model	Conf.	Acc	ECE	Conf|corr.	Conf|wrong
GPT-5.5	83.9	0.463	0.376	83.1	**84.5**
Claude Opus 4.7	82.3	0.498	0.325	84.4	80.2
Gemini 3.1 Pro	95.2	0.515	0.438	94.7	**95.7**

responses, and commitment is itself selective, so the committed subset may differ systematically in difficulty across strategies. Testing the mechanism would require manipulating the assertion threshold explicitly, for example by varying an instructed decision criterion while holding the scaffold fixed.

Both models commit over wide ranges, 0.271–0.903 for GPT-5.5 and 0.634–0.953 for Claude, and occupy different regions of that range. That difference offers a plausible reading of the opposed rankings in Section 4.1: a scaffold that suppresses commitment moves GPT-5.5 away from its observed optimum and Claude toward its own. We offer this as an interpretation the data support rather than as a demonstrated cause.

4.4 Confidence is not a usable trust signal (RQ3)

On self-consistency, the one strategy for which all three models used an identical output schema, accuracies on the 717 videos all three completed were GPT-5.5 0.471, Claude 0.498, and Gemini 0.515 (GPT-5.5 attains 0.463 on its full 812-video run, as in Tables 1 and 4). Cochran's $Q = 6.08$ ($p = 0.048$); only the GPT–Gemini pair separates under McNemar's exact test ($p = 0.014$). A 4.4-point spread across three frontier models is small beside the 17–50-point spread across strategies within a model.

Calibration is poor in all three (Table 4), and the direction of the failure differs. Claude's stated confidence is at least ordered correctly: 84.4 when right against 80.2 when wrong, giving a positive confidence–correctness correlation ($r = +0.221$, $p < 0.001$). GPT-5.5 inverts, reporting *higher* confidence on the answers it gets wrong (84.5 against 83.1, $r = -0.089$, $p = 0.011$). Gemini is the extreme case, averaging 95.2% confidence at 51.5% accuracy, again with higher confidence on incorrect responses.

Agreement across sampled generations behaves better than the stated number: under GPT-5.5, inter-sample agreement correlates with correctness at $r = +0.22$, and accuracy in the highest-agreement tercile reaches 0.572 against 0.390 in the lowest. The signal that is free is informative; the signal that is requested is not.

4.5 Error structure: taxonomy boundaries versus perception

Not every misclassification is a perceptual failure. UCF-Crime assigns one label per video, but surveillance footage routinely satisfies several category definitions at once: a shove followed by a snatched bag is simultaneously *Abuse*, *Assault*, and *Robbery*. To separate boundary disputes from perceptual error we group the eleven categories into four semantic families (theft-related, violence-related, destruction-related, and accidents) and rescore at family level.

Table 5: Per-class recall pooled across strategies, with test-split support.

Category	GPT-5.5	Claude	n
Explosion	**0.634**	0.598	50
Assault	0.548	0.562	12
RoadAccidents	0.508	0.553	150
Burglary	0.506	0.649	100
Robbery	0.432	**0.767**	150
Stealing	0.360	0.381	100
Fighting	0.349	0.222	50
Shoplifting	0.314	0.420	50
Vandalism	0.257	0.140	50
Abuse	0.231	0.455	50
Shooting	0.083	0.149	50

Family-level accuracy exceeds exact-label accuracy by 0.103 for GPT-5.5 (0.409 to 0.512) and 0.136 for Claude (0.530 to 0.666). Among Claude's errors, 28.8% assign a label within the correct family, predominantly *Stealing*→*Burglary* (158 cases) and *Fighting*→*Assault* (142), against 17.5% for GPT-5.5, whose errors are dominated instead by abstention (59.4% resolve to *Normal*). The two models therefore fail differently in a second sense: Claude disputes the boundary, GPT-5.5 declines the question.

This matters for interpreting absolute accuracy. A non-trivial share of measured error reflects a single-label taxonomy applied to multi-event footage rather than failure to perceive the event, which argues for multi-label or family-level reporting in forensic evaluation.

4.6 Class structure and the perceptual floor

Table 5 reports per-class recall pooled across strategies. The ordering is broadly shared between models: *Explosion* and *RoadAccidents* are recovered reliably, *Shooting* almost never. The distinguishing property is temporal extent. Explosions and vehicle collisions leave visual traces (smoke, debris, stopped traffic) that persist across many sampled frames, whereas a discharge event may occupy a fraction of a second and fall between samples entirely.

Recall on *Shooting* is at or below 0.204 for *every* model–strategy pair (GPT-5.5 0.000–0.160; Claude 0.102–0.204), against *Explosion* at 0.292–0.840 and 0.520–0.700. No reorganisation of reasoning recovers evidence that frame sampling has discarded. This bounds the contribution available from prompt engineering and locates the residual error in the input pipeline. Where the two models diverge sharply, as with *Robbery* at 0.432 versus 0.767 and *Abuse* at 0.231 versus 0.455, the gap tracks each model's commitment threshold rather than any evident perceptual difference.

4.7 Justification language and error (RQ3)

A forensic triage aid does not only emit a label; it emits the evidence it claims to rely on, and that text is what an investigator reads. We therefore audit the stated supporting evidence for descriptors of persons.

Table 6: Relative risk of misclassification given marker presence in the stated supporting evidence. Rates are averaged over strategies.

Model	Marker class	Rate	RR (95% CI)
GPT-5.5	Status	0.065	**1.21** (1.14–1.28)
	Apparel	0.087	**0.59** (0.53–0.66)
	"Suspicious person"	0.044	1.02 (0.93–1.11)$^{\text{n.s.}}$
Claude	Status	0.160	**1.39** (1.31–1.48)
	Apparel	0.174	**0.76** (0.70–0.83)
	"Suspicious person"	0.030	1.36 (1.21–1.53)

Explicit demographic descriptors anchored to a person are rare in both models. Two proxy classes are far more common, and they behave in opposite directions (Table 6).

Status markers (homelessness, vagrancy, panhandling, dishevelment, intoxication, loitering) are associated with *elevated* error: relative risk 1.21 (95% CI 1.14–1.28) for GPT-5.5 and 1.39 (1.31–1.48) for Claude, both $p < 10^{-10}$. *Apparel* markers (hooded or masked figures, obscured faces) are associated with *reduced* error: RR 0.59 (0.53–0.66) and 0.76 (0.70–0.83).

These are associations, not causal effects: we do not claim that invoking a status descriptor causes the misclassification. Both could follow from a third factor, for instance footage in which the depicted activity is genuinely ambiguous, which might elicit status language and independently make the category harder to name. The association is nonetheless notable for two reasons. First, the sign reverses between the two marker classes, so the pattern is not an artefact of descriptor counting: concealed faces are genuine tradecraft that legitimately raise the posterior on robbery, and citing them is associated with correct classification, whereas housing status and intoxication are not evidence of the specific offence under adjudication and citing them is associated with error. Second, two independently developed models reproduce both associations and both signs. The generic phrase "suspicious individual" is associated with error under Claude (RR 1.36) but not under GPT-5.5 (RR 1.02, $p = 0.74$), suggesting the pattern is tied to status content rather than to evaluative language in general.

Whatever its origin, nothing in the models' own confidence distinguishes the two cases. Combined with the calibration results of Section 4.4, a triage output can cite social status as supporting evidence, assign high confidence, and belong to a subset with above-average error, with no signal available to the reader that any of this has occurred.

4.8 Cost (RQ3)

Strategies differ by more than an order of magnitude in generated output (Table 7), and cost is not aligned with accuracy in either model. For GPT-5.5 the cheapest strategy is also the most accurate: zero-shot produces 1,497 tokens per video at 0.596 accuracy, against self-consistency's 13,200 at 0.463. Normalised as tokens per correct classification the gap is 2,511 against 28,506, an 11-fold premium for a 0.133 loss.

Claude's ordering differs again. Its best strategy, least-to-most, is also its most expensive at 59,425 tokens per video, while sequential attains almost the same accuracy (0.599 against 0.616) for a tenth of the output. Iterative refinement is the worst value in the study at 177,053 tokens per correct classification, though this run covers only 432 videos and the figure should be read with that in mind. Because neither the accuracy ordering (Section 4.1) nor the cost ordering transfers across models, a deployment cannot import a cost–accuracy trade-off from published single-model results.

Table 7: Generation cost. Tokens are mean generated output per video, estimated from character counts.

Model	Strategy	Tok./video	Acc	Tok./correct
GPT-5.5	Zero-shot	**1,497**	**0.596**	**2,511**
	Sequential	4,790	0.482	9,947
	Self-consistency	13,200	0.463	28,506
	Meta-prompting	3,466	0.443	7,825
	ReAct	8,005	0.420	19,061
	Iterative	2,866	0.358	7,996
	Least-to-most	4,313	0.096	44,904
Claude	Least-to-most	59,425	**0.616**	96,470
	Sequential	**5,776**	0.599	**9,643**
	Chain-of-thought	6,353	0.527	12,053
	ReAct	12,780	0.523	24,417
	Meta-prompting	14,325	0.514	27,894
	Self-consistency	14,389	0.498	28,898
	Zero-shot	4,860	0.498	9,768
	Iterative	78,690	0.444	177,053

5 Discussion

Answers to the research questions. **RQ1**: strategy rankings do not transfer. No strategy was optimal for all three models, and two moved from first place to seventh between GPT-5.5 and Claude. **RQ2**: the variation is largely attributable to commitment behaviour. Commitment rate tracks accuracy at $r = 0.964$ and 0.806, while conditional accuracy holds within 0.660–0.694 across GPT-5.5's five mainstream strategies and is orthogonal to accuracy under Claude; a minority of scaffolds have lower values on both components. We read this as consistent with prompting acting primarily on assertion behaviour, without claiming a demonstrated causal mechanism. **RQ3**: practitioners should select a strategy per model rather than from published rankings, treat verbalised confidence with caution and prefer sampling agreement where multiple generations are affordable, treat commitment rate as an explicit policy setting rather than an accident of prompt design, and audit justification text alongside labels. The remainder of this section develops these four consequences.

Strategy effects are properties of the strategy–model pair. The practical consequence is direct: a practitioner who adopts published prompting guidance derived from one model has no evidential basis for expecting it to hold on another. In our data the guidance would have been actively harmful in both directions: zero-shot for Claude costs 0.118 accuracy against that model's best strategy, and least-to-most for GPT-5.5 costs 0.500.

Reconciling with rubric-based evaluation. Our prior work using a ten-criterion forensic rubric found ReAct strongest for all three

models tested [14]; here ReAct ranks fifth of seven for GPT-5.5 and fourth of eight for Claude. The measures differ: a rubric rewards temporal reconstruction, evidence cataloguing, and court-ready narrative, all output properties that ReAct produces by construction, whereas classification correctness asks whether the named category is right. A well-structured, thoroughly evidenced narrative about the wrong crime scores highly on the first and zero on the second. For forensic triage this divergence matters: the output most likely to persuade an investigator is not necessarily the output most likely to be correct.

Evidence language is part of the output. Section 4.7 shows that the descriptors a model cites carry information about whether it is right, and that the informative direction is the opposite of what a careless reader would assume: concealment cues track correctness, social-status cues track error. Because neither is distinguished by the model's stated confidence, and because confidence is itself anti-correlated with correctness in two of three models, a reviewer of these outputs has no available signal for the distinction. We suggest that forensic triage systems be evaluated on their justification text as well as their labels, and that status descriptors be treated as a flag for review rather than as supporting evidence.

Implications for forensic deployment. Commitment rate is not a neutral parameter. A configuration with low commitment declines to name a crime on footage that contains one; a configuration with high commitment names a specific crime, correctly for roughly two-thirds of committed responses and incorrectly for the remainder, against people recorded in surveillance footage. Our evaluation partition contains no normal videos, so we cannot measure the false-alarm rate on genuinely uneventful footage and cannot quantify the trade-off between missed incidents and false accusations as it would appear in deployment. What our results do show is that prompt structure shifts the balance between abstention and incorrect assertion of a specific crime, and that this operating point is currently set implicitly by prompt design rather than deliberately by policy. We recommend that deployments report commitment rate and conditional accuracy separately, treat the threshold as an explicit configurable, and disregard verbalised confidence in favour of sampling agreement where multiple generations are affordable. Because the accuracy ordering does not transfer, neither does the cost ordering (Section 4.8): the most expensive strategy we evaluated is also the least accurate for one of the two models on which cost could be measured.

6 Limitations

Task scope. All results come from a single task and benchmark. We cannot exclude that properties of UCF-Crime, such as its single-label taxonomy, the absence of *Normal* videos, or the brevity of some events relative to frame sampling, contribute to the inversions we observe. The claim we support is accordingly narrower than a general one: for this task, a strategy ranking derived from one frontier model did not predict the ranking on another. Whether the same model dependence holds for structurally different multimodal tasks, such as document understanding, chart reasoning, or medical imaging, remains open, and a smaller replication on two or three such tasks is the most direct test.

Evidentiary scope. We evaluate candidate methods, not a deployable forensic system. Chain of custody, authentication, data residency, and admissibility requirements are unaddressed here and must be satisfied before any operational use.

No *Normal* videos. The evaluated partition contains only anomaly categories, so every normal-scene response is scored incorrect and no false-alarm rate is available. Results are therefore anomaly *classification*, not detection.

Coverage. Four Claude runs were truncated by API rate limiting (sequential, self-consistency, least-to-most, and iterative refinement), in one case missing-not-at-random. Where affected we report matched-subset results alongside full-run figures.

Determinism. GPT-5.5 and Claude runs at low temperature were reproducible to the prediction; only self-consistency was stochastic. We therefore report no seed-level variance estimate.

Harness asymmetry. Gemini's harness elicited prose for seven of the eight strategies, so that model enters the direct analysis only through self-consistency, the one strategy using the shared output schema, and otherwise through the panel measure. Its position in the technique ranking therefore rests on the coarser measure.

Statistical power over strategies. With eight strategies, rank correlations are underpowered; we accordingly rest RQ1 on specific inversions.

Associations, not mechanisms. The commitment decomposition is an accounting identity and the marker analysis is observational. Neither establishes that a scaffold acts on the assertion threshold, nor that a descriptor class causes misclassification. Conditional accuracy is computed on a selected subset (the committed responses) and is therefore not a clean measure of perceptual ability.

Panel scores are not cross-model estimates. Because self-judging was excluded, each subject model is assessed by a different pair of judges of differing severity. We use panel results for within-model strategy orderings, not for ranking the models against one another.

Marker detection is lexical. Justification markers are identified by person-anchored regular expressions, not by human annotation. The patterns, with their precision and recall, are available from the authors on request. The analysis speaks to descriptor use in generated text; because UCF-Crime carries no demographic annotation, we make no claim about disparate error rates across demographic groups, and such a claim would require data we do not have.

Cost is a proxy. Tokens are estimated from generated character counts rather than logged API usage, and are unavailable for Gemini.

Cultural and jurisdictional specificity. UCF-Crime is US-sourced and its taxonomy reflects US law-enforcement categories. Constructs such as *Abuse*, *Arrest*, and *Fighting* are legally and culturally contingent, and neither our results nor the benchmark should be assumed to transfer to other jurisdictions.

7 Conclusion

Across eight prompting strategies, three frontier multimodal models, and 807 forensic videos, no strategy was optimal for all models, and individual strategies inverted by up to six rank positions and 0.520 accuracy between model families. The variation is carried almost entirely by how readily a model commits to asserting a crime

rather than by how well it discriminates once committed, which explains why an identical scaffold helps one model and harms another. Verbalised confidence was anti-correlated with correctness in two of three models, while sampling agreement was informative. Prompting guidance derived from a single model does not license cross-model recommendation. The three questions we set out with resolve as follows: rankings do not transfer; most of the variation is associated with how readily a model commits to a crime label rather than with its accuracy once committed; and in forensic triage that commitment behaviour governs the balance between declining to name a crime and naming the wrong one.

Two extensions would sharpen these findings. First, re-running all eight strategies under a shared structured-output schema for all three models would extend the direct analysis, including the commitment decomposition, to Gemini, and would replace the single matched-strategy comparison of Section 4.1 with a full between-model estimate. Second, expert human labelling of a stratified subset would give a reference against which both direct parsing and the judge panel could be validated, and would remove the judge-severity imbalance described in Section 3.5.

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